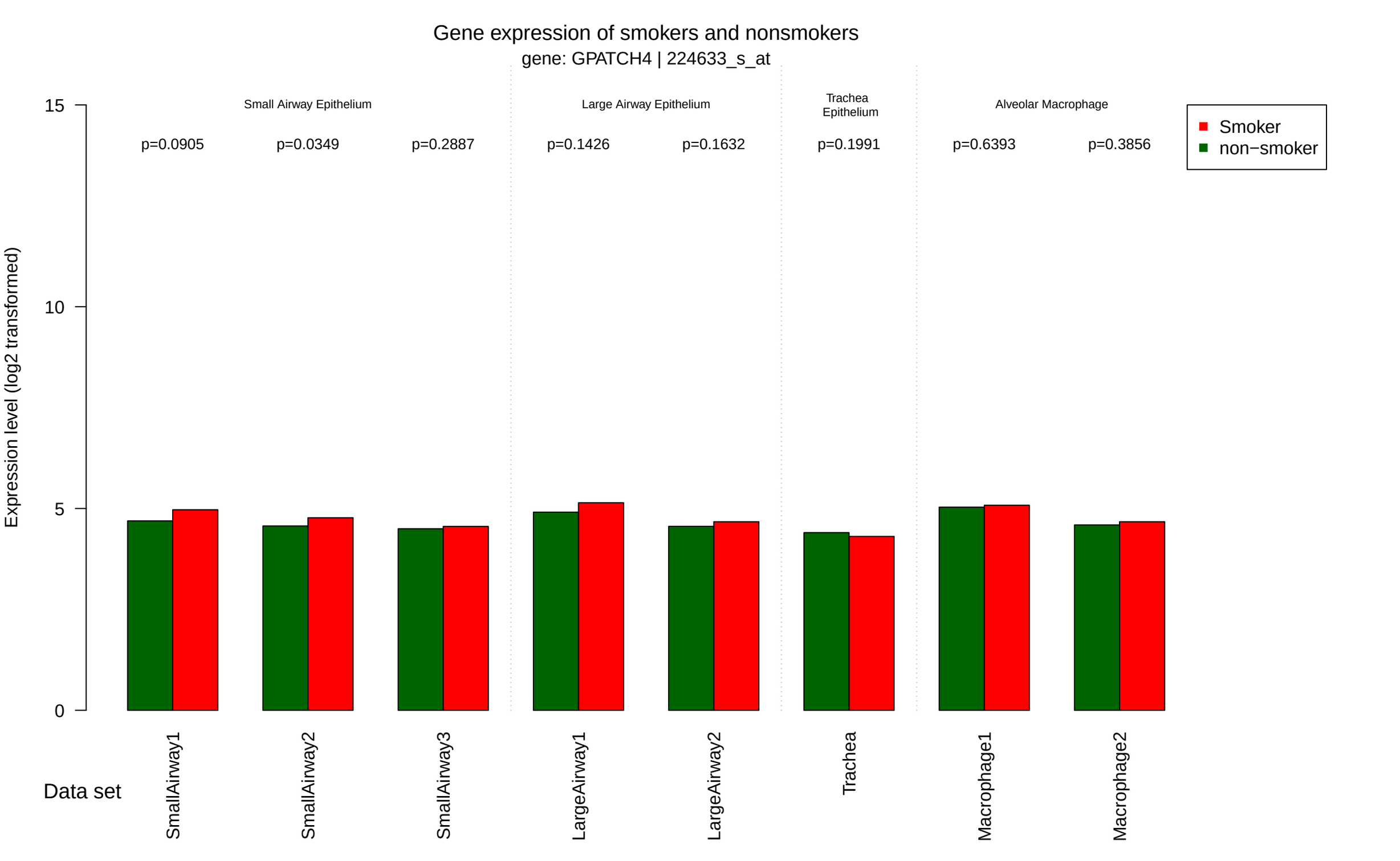
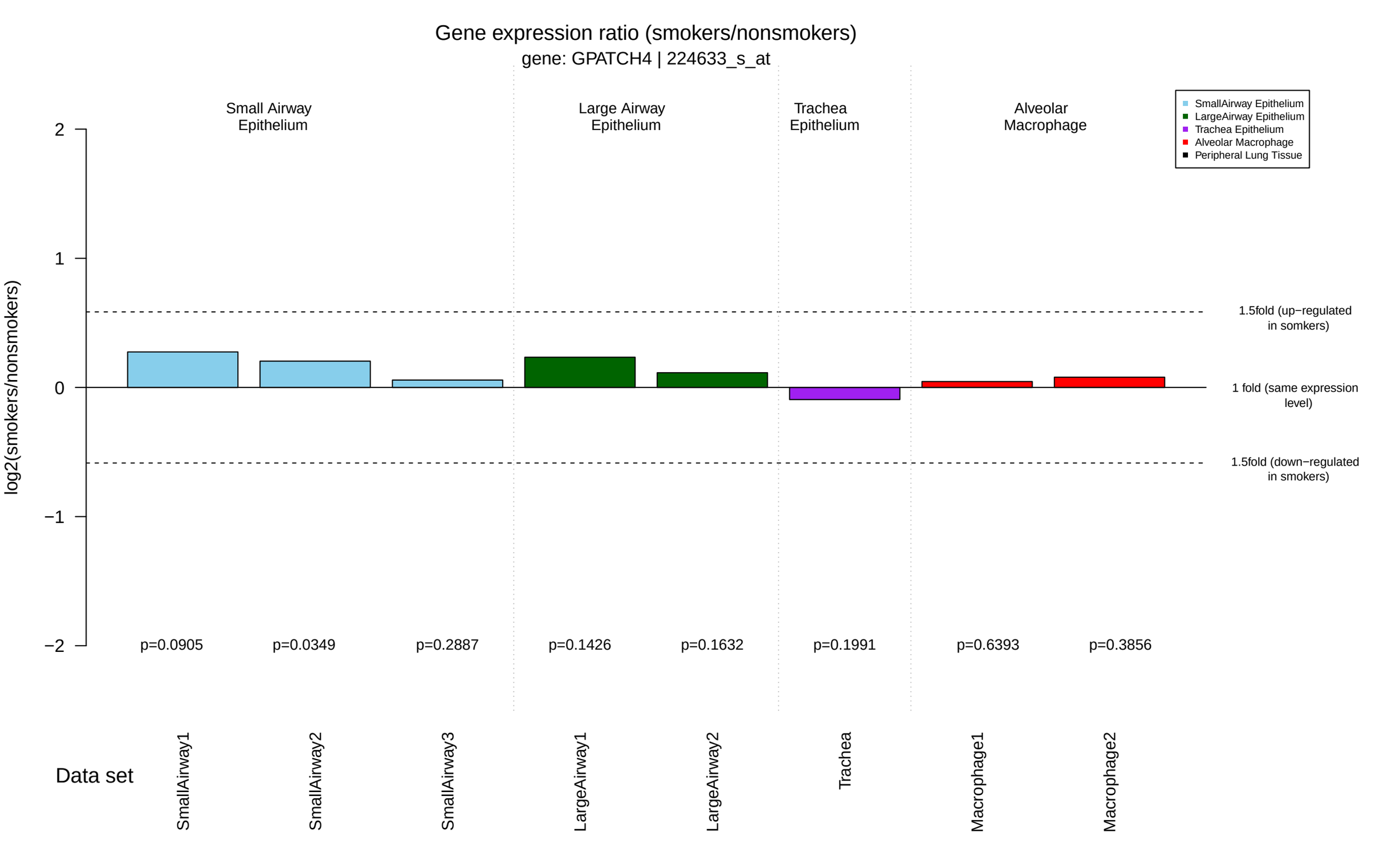
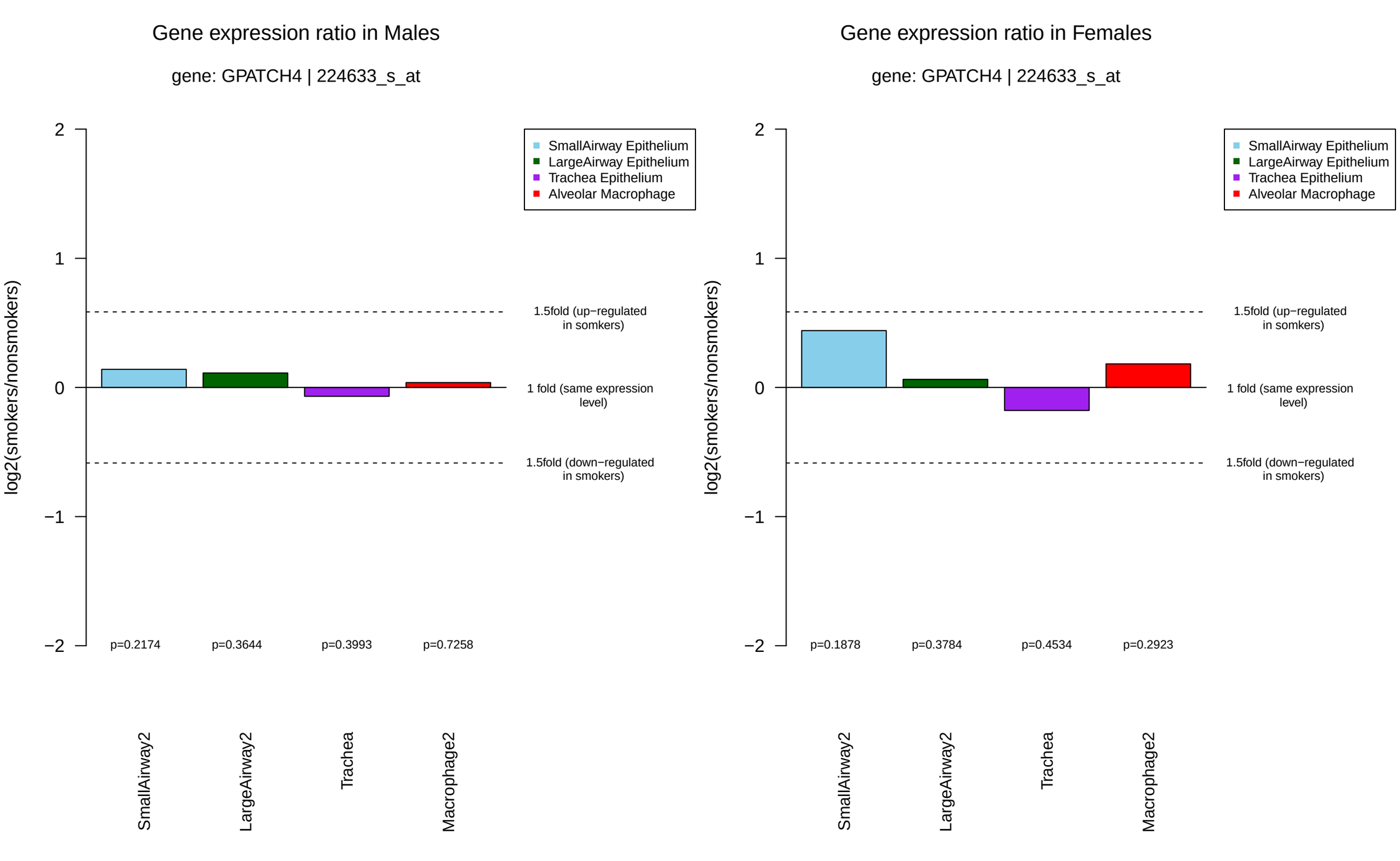
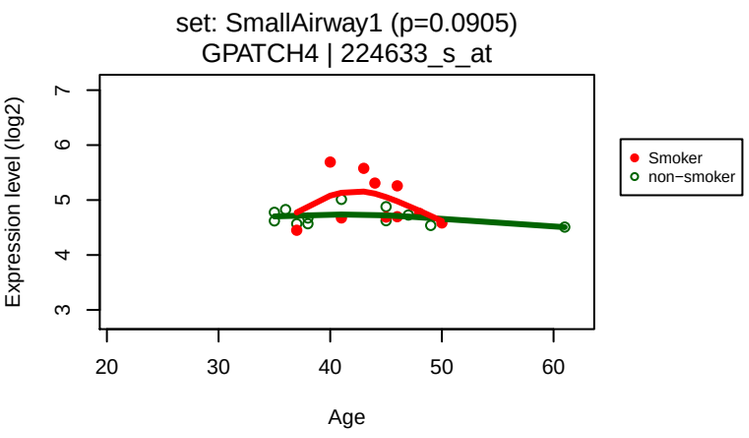
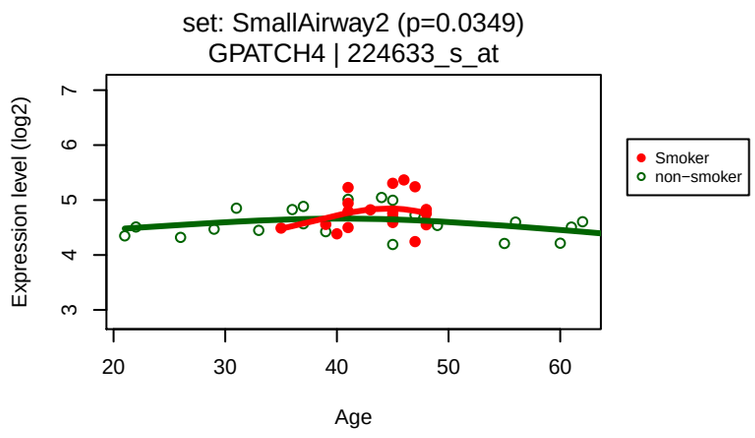
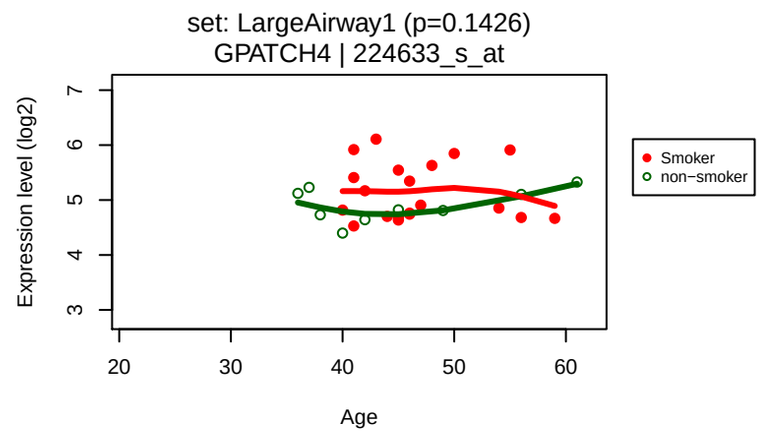

Gene expression of smokers and nonsmokers
gene: GPATCH4 | 224633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0905
p=0.0349
p=0.2887
p=0.1426
p=0.1632
p=0.1991
p=0.6393
p=0.3856
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

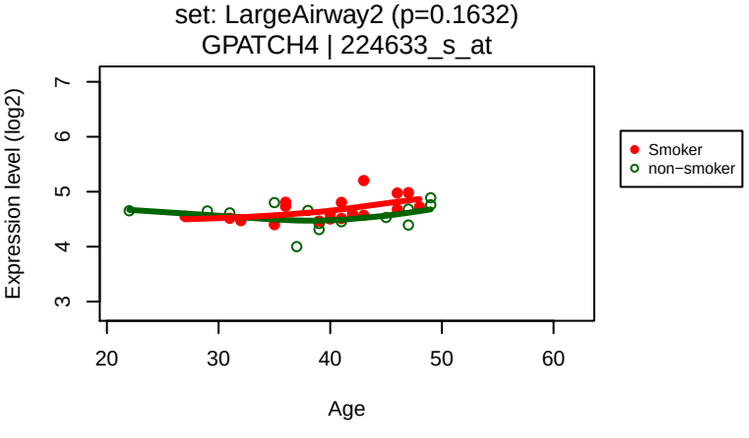

Gene expression ratio (smokers/nonsmokers)
gene: GPATCH4 | 224633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0905
p=0.0349
p=0.2887
p=0.1426
p=0.1632
p=0.1991
p=0.6393
p=0.3856
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

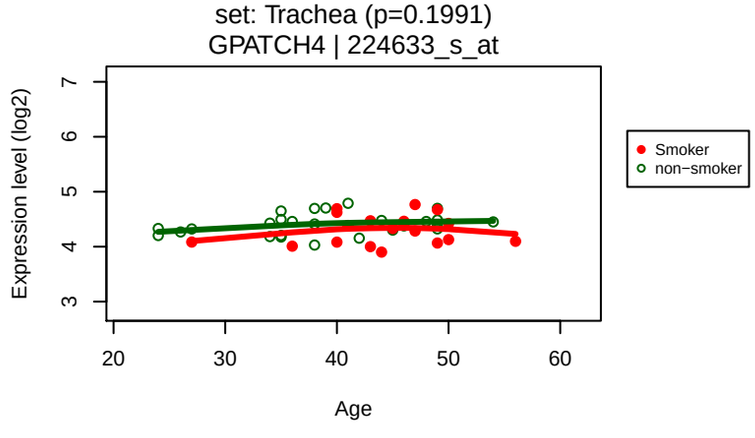

Gene expression ratio in Males
gene: GPATCH4 | 224633_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2174
p=0.3644
p=0.3993
p=0.7258
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPATCH4 | 224633_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1878
p=0.3784
p=0.4534
p=0.2923
SmallAirway2
LargeAirway2
Trachea
Macrophage2

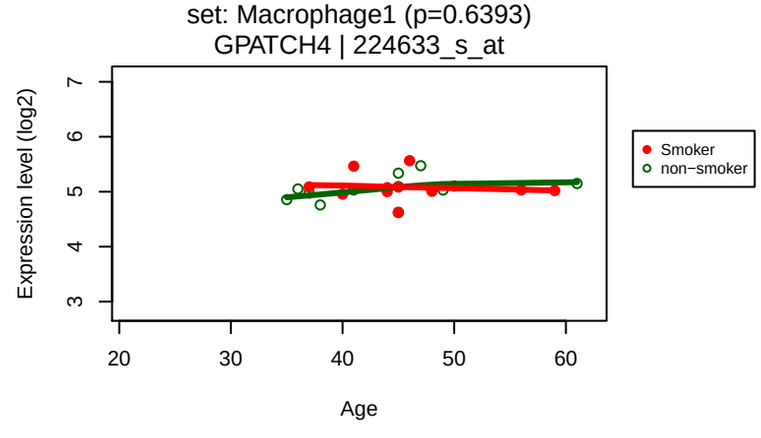

set: SmallAirway1 (p=0.0905)
GPATCH4 | 224633_s_at
Expression level (log2)
Age
Smoker
non−smoker

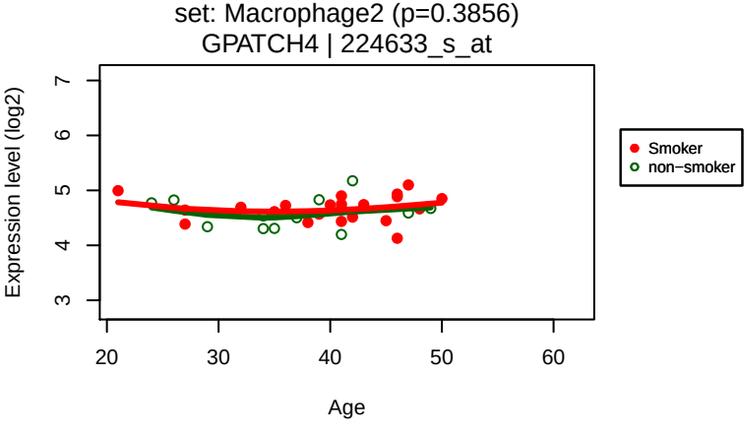

set: SmallAirway2 (p=0.0349)
GPATCH4 | 224633_s_at
Expression level (log2)
Age
Smoker
non−smoker

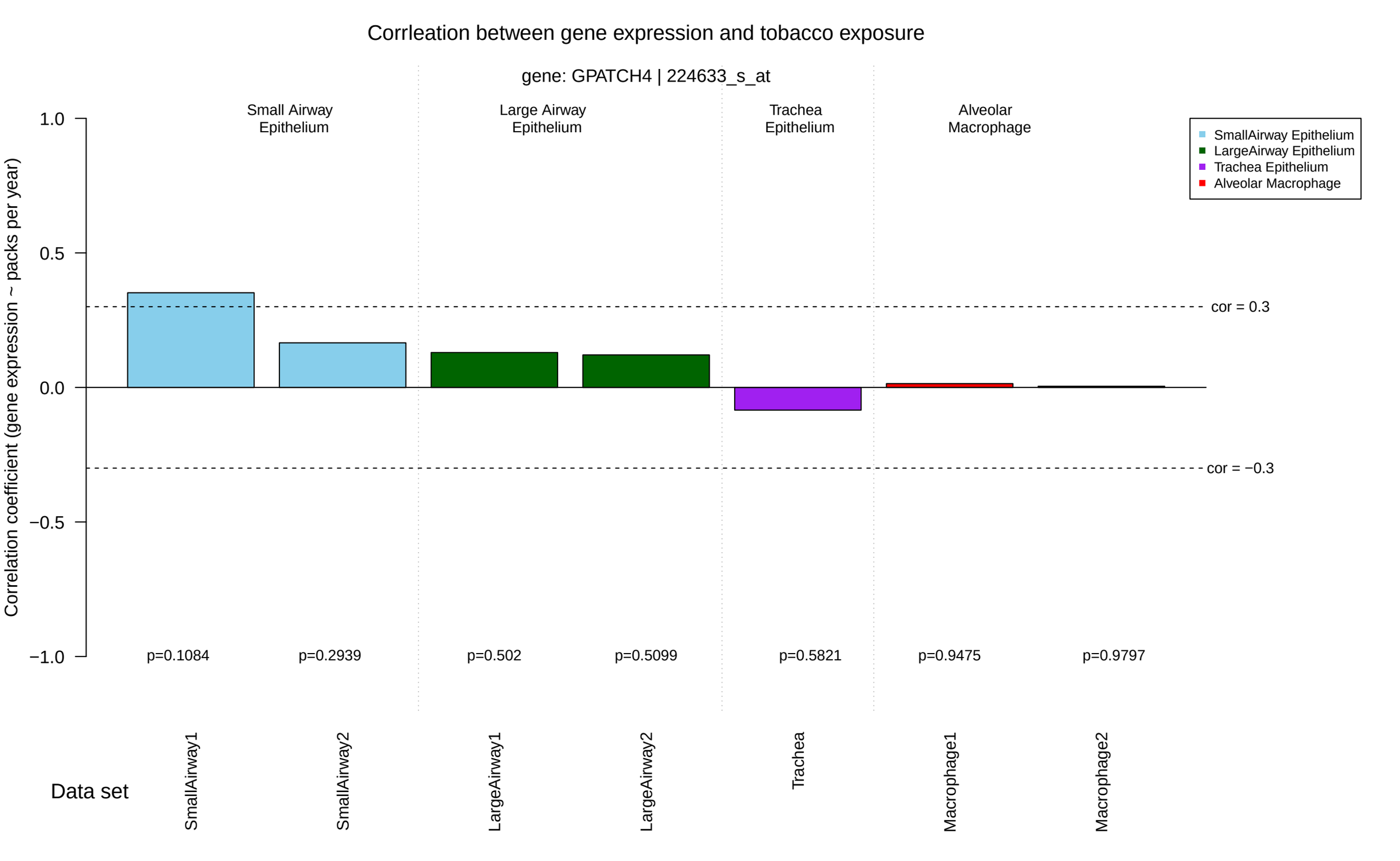

set: LargeAirway1 (p=0.1426)
GPATCH4 | 224633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1632)
GPATCH4 | 224633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1991)
GPATCH4 | 224633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6393)
GPATCH4 | 224633_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3856)
GPATCH4 | 224633_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPATCH4 | 224633_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1084
p=0.2939
p=0.502
p=0.5099
p=0.5821
p=0.9475
p=0.9797
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2